CS 182 - HW 13 - Jie Wen

1. DDPM/DDIM Fun: From a Gaussian

(a) Distribution of X_1

Solution: The total noise added over $[0, 1]$ has variance 1.

$$x_1 \sim \mathcal{N}(0, \sigma^2 + 1) \approx \mathcal{N}(0, 1) \text{ since } \sigma^2 \ll 1$$

This is because we have T different independent Gaussians of noise being added together, each of variance $\Delta t = \frac{1}{T}$.

(b) Marginals of $X_{t-\Delta t}$ and X_t

Solution: These are obtainable from the same argument as the previous part, using t instead of 1.

$$X_t \sim \mathcal{N}(0, \sigma^2 + t)$$

$$X_{t-\Delta t} \sim \mathcal{N}(0, \sigma^2 + t - \Delta t)$$

We can think of this as tT steps of adding independent Gaussian noise of variance $\Delta t = \frac{1}{T}$. Doing one step less results in Δt less variance.

(c) Conditional Variance Approximation

Solution:

$$Var(x_{t-\Delta t}|x_t) = \frac{(\sigma^2 + t - \Delta t)\Delta t}{\sigma^2 + t} = \Delta t(1 - \frac{\Delta t}{\sigma^2 + t}) \approx \Delta t$$

(d) Naive Reverse Denoising (No Noise)

Solution: Think of this as T steps working backward from $k = T - 1$ to $k = 0$. The factor applied at step k is $\alpha_k = \frac{\sigma^2 + k\Delta t}{\sigma^2 + (k+1)\Delta t}$.
Taking the product of these factors results in telescoping:

$$\prod_{k=0}^{T-1} \alpha_k = \frac{\sigma^2}{\sigma^2 + T\Delta t} = \frac{\sigma^2}{\sigma^2 + 1}$$

The resulting distribution is:

$$\hat{X}_0 = (\frac{\sigma^2}{\sigma^2 + 1})X_1 \Rightarrow \hat{X}_0 \sim \mathcal{N}\left(0, \left(\frac{\sigma^2}{\sigma^2 + 1}\right)^2\right) \approx \mathcal{N}(0, \sigma^4)$$

Since $\sigma^2 \ll 1$, the variance is far too small (approximately zero).

(e) Reverse Denoising with Noise (Limit)

Solution: Treating Δt as dt and viewing the variance sum as a Riemann sum:

$$Var(\hat{X}_0) \approx \left(\frac{\sigma^2}{\sigma^2 + 1}\right)^2 + \int_0^1 \left(\frac{\sigma^2}{\sigma^2 + t}\right)^2 dt$$

Evaluating the integral:

$$= \sigma^4\left(\frac{1}{\sigma^2 + 1}\right)^2 + \sigma^4\left(\frac{1}{\sigma^2} - \frac{1}{\sigma^2 + 1}\right)$$

$$= \sigma^2 + \sigma^4\left(\frac{1}{(\sigma^2 + 1)^2} - \frac{1}{\sigma^2 + 1}\right) \approx \sigma^2$$

The final approximation holds because $\sigma^6 \ll \sigma^2$, thus recovering the desired variance.

(f) Training a Neural Network for Estimation

Solution:

- **Inputs:** The position $t \in [0, 1]$ and the noisy sample x_t.
- **Batch Generation:**
  1. Draw k iid copies for t_i from a uniform random variable on $[0, 1]$.
  2. Draw Gaussian noise z_i from $\mathcal{N}(0, t_i)$.
  3. Draw k random training points s_{l_i} from the dataset.
  4. Define input pairs as $(t_i, s_{l_i} + z_i)$.
- **Loss Computation:** Use the squared loss trying to predict the original sample from the noisy version:

$$(s_{l_i} - f_\theta(t_i, s_{l_i} + z_i))^2$$

(g) DDIM Step Approximation

Solution:

$$x_{t-\Delta t}^{DDIM} = (1 - \frac{\eta(t, \Delta t)\Delta t}{\sigma^2 + t})x_t$$

Using the expansion $\eta(t, \Delta t) \approx \frac{1}{2} + \frac{\Delta t}{8t}$:

$$x_{t-\Delta t}^{DDIM} \approx x_t \left(1 - \frac{\Delta t}{2(\sigma^2 + t)} - \frac{\Delta t^2}{8t(\sigma^2 + t)}\right)$$

This simplifies to approximately $x_t(1 - \frac{\Delta t}{2(\sigma^2+t)})$.

(h) DDIM Total Contraction

Solution:

$$\hat{X}_0 \approx X_1 \cdot \prod_{k=1}^{T} \left(1 - \frac{\Delta t}{2(\sigma^2 + k\Delta t)}\right)$$

Taking logs and limits (using $\ln(1 - x) \approx -x$):

$$\ln \left(\prod_{k=1}^{T} \left(1 - \frac{\Delta t}{2(\sigma^2 + k\Delta t)}\right)\right)^2 \approx 2 \sum_{k=1}^{T} - \frac{\Delta t}{2(\sigma^2 + k\Delta t)}$$

Approximating as an integral:

$$\approx -2 \int_0^1 \frac{1}{2(\sigma^2 + t)} dt = -(\ln(\sigma^2 + 1) - \ln \sigma^2) = \ln \frac{\sigma^2}{\sigma^2 + 1}$$

Thus:

$$\hat{X}_0 = X_1 \cdot \left(\frac{\sigma^2}{\sigma^2 + 1}\right)^{1/2} \Rightarrow \hat{X}_0 \sim \mathcal{N}\left(0, \frac{\sigma^2}{\sigma^2 + 1}\right) \approx \mathcal{N}(0, \sigma^2)$$

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2. Honey, Where's My Reward Model?

(a) Optimal p^*

Solution: Since the KL divergence is non-negative ($\mathbb{D}_{KL}(p||q) \geq 0$) and equals zero if and only if $p(x) = q(x)$ for all x, the solution is $p^* = q$.

(b) Optimal Policy Form

Solution: Expanding the objective function with respect to π_θ:

$$min_{\pi_\theta} \mathbb{E}_{x \sim \mathcal{D}} \left[\mathbb{D}_{KL}\left(\pi_\theta(y|x)||\frac{1}{Z(x)}\pi_{ref}(y|x) \exp\left(\frac{1}{\beta} r_\phi(x, y)\right)\right) - \log(Z(x))\right]$$

The partition function $Z(x)$ is defined as $\sum_y \pi_{ref}(y|x) \exp(\frac{1}{\beta} r_\phi(x, y))$.
The optimal policy is:

$$\pi_{\theta^*}(y|x) = \frac{1}{Z(x)}\pi_{ref}(y|x) \exp\left(\frac{1}{\beta} r_\phi(x, y)\right)$$

This uses the fact that minimizing the KL divergence leads to the distributions being equal.

(c) Difficulty of $Z(x)$

Solution: $Z(x)$ requires summing over all possible outputs $y \in \mathcal{Y}$. In practice, this set is astronomically large (exponential with sequence length L). For example, with GPT-3, the output space size can be approx 1.1×10^{9628}, making exact computation impossible.

(d) Expressing $r_\phi(x, y)$

Solution: Starting with the optimal policy expression and taking logarithms:

$$\log \pi_{\theta^*}(y|x) = -\log Z(x) + \log \pi_{ref}(y|x) + \frac{1}{\beta} r_\phi(x, y)$$

Rearranging for $r_\phi(x, y)$:

$$r_\phi(x, y) = \beta \log \left(\frac{\pi_{\theta^*}(y|x)}{\pi_{ref}(y|x)}\right) + \beta \log Z(x)$$

(e) DPO Derivation (Bradley-Terry substitution)

Solution: Substituting the reward expression into the Bradley-Terry model:

$$p_{\theta^*}(y_1 > y_2|x) = \frac{1}{1 + \exp(\beta \log(\frac{\pi^*(y_2|x)}{\pi_{ref}(y_2|x)}) - \beta \log(\frac{\pi^*(y_1|x)}{\pi_{ref}(y_1|x)}))}$$

$$= \sigma\left(\beta \log \left(\frac{\pi^*(y_1|x)}{\pi_{ref}(y_1|x)}\right) - \beta \log \left(\frac{\pi^*(y_2|x)}{\pi_{ref}(y_2|x)}\right)\right)$$

The partition function $Z(x)$ cancels out, making computation feasible. The final Maximum Likelihood objective (DPO Loss) is:

$$\mathcal{L}_{DPO}(\pi_\theta; \pi_{ref}) = -\mathbb{E}_{(x,y_w,y_l)\sim\mathcal{D}}\left[\log\sigma\left(\beta\log\frac{\pi_\theta(y_w|x)}{\pi_{ref}(y_w|x)} - \beta\log\frac{\pi_\theta(y_l|x)}{\pi_{ref}(y_l|x)}\right)\right]$$

(f) DPO Gradient and Interpretation

Solution: The gradient is derived as:

$$\nabla_\theta\mathcal{L}_{DPO} = -\mathbb{E}_{(x,y_w,y_l)\sim\mathcal{D}}\left[\beta\sigma(\hat{r}_\theta(x,y_l) - \hat{r}_\theta(x,y_w))[\nabla_\theta\log\pi(y_w|x) - \nabla_\theta\log\pi(y_l|x)]\right]$$

Interpretation:

- The term $\nabla_\theta\log\pi(y_w) - \nabla_\theta\log\pi(y_l)$ pushes the model to increase the likelihood of the winner and decrease the loser.
- The scalar weight $\sigma(\hat{r}_\theta(x,y_l) - \hat{r}_\theta(x,y_w))$ acts dynamically:
  - If the model is correct (winner reward $\gg$ loser reward), the weight approaches 0 (gradient vanishes).
  - If the model is incorrect (loser reward $>$ winner reward), the weight approaches 1 (strong gradient to correct the mistake).

(g) Plackett-Luce Model

Solution: Using the reparameterization $exp(r_\phi(x,y)) = exp(\beta\log\frac{\pi_{\theta^*}(y|x)}{\pi_{ref}(y|x)}) \cdot Z(x)^\beta$.
When substituted into the Plackett-Luce product for the probability of a ranking τ:

$$p_{\theta^*}(\tau|y_1,\ldots,y_K,x) = \prod_{k=1}^{K}\frac{Z(x)^\beta\cdot\exp(\beta\log\frac{\pi_{\theta^*}(y_{\tau(k)}|x)}{\pi_{ref}(y_{\tau(k)}|x)})}{\sum_{j=k}^{K}Z(x)^\beta\cdot\exp(\beta\log\frac{\pi_{\theta^*}(y_{\tau(j)}|x)}{\pi_{ref}(y_{\tau(j)}|x)})}$$

The term $Z(x)^\beta$ factors out of the sum in the denominator and cancels with the numerator completely.

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4. Diffusion Models (Exam Preparation)

(a) Anytime Sampling from Intermediate Distributions

Solution: Using the reparameterization trick $x = \mu + \sigma\epsilon$:

$$x_t = \sqrt{\gamma_t}\sqrt{\gamma_{t-1}}x_{t-2} + \sqrt{1-\gamma_t\gamma_{t-1}}\hat{\epsilon}_{t-2}$$

By unrolling the recursion and defining $\alpha_t = \prod_{i=1}^{t}\gamma_i$:

$$q(x_t|x_0) = \mathcal{N}(x_t; \sqrt{\alpha_t}x_0, (1-\alpha_t)I)$$

(b) Reversing the Diffusion Process

Solution: Applying Bayes rule: $q(x_{t-1}|x_t,x_0) \propto q(x_t|x_{t-1},x_0)q(x_{t-1}|x_0)$.
By plugging in the Gaussian densities and completing the square in the exponent, we recover the mean and variance:

$$\hat{\beta}_t = \frac{1-\alpha_{t-1}}{1-\alpha_t}\beta_t$$

$$\mu(x_t,x_0) = \frac{\sqrt{1-\beta_t}(1-\alpha_{t-1})}{1-\alpha_t}x_t + \frac{\beta_t\sqrt{\alpha_{t-1}}}{1-\alpha_t}x_0$$

Thus $q(x_{t-1}|x_t,x_0) \sim \mathcal{N}(\mu(x_t,x_0), \hat{\beta}_t I)$.